

W3L1 | Lens detective

Monday | 09:30–10:10 | Introduce | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Total magnification = eyepiece $\times$ objective. A bigger image is not always a clearer image. Describe what you can see before you explain it.

Work together

Choose $\times 4$ then $\times 40$ objectives with a $\times 10$ eyepiece. Predict the total each time.

1. A $\times 10$ eyepiece is used with $\times 4$, $\times 10$ and $\times 40$ objectives. Calculate the three totals.

2. A blurred image is enlarged. Does it necessarily show two close dots more clearly? Explain.

W3L1 | Continue and improve

Use after the core questions when directed

3. **Record your route: real microscope / digital model. Give one observation and one thing this route cannot establish.**

Optional transfer

Why must the specimen be centred before increasing magnification?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W3L2 | Calculation relay — without a race

Wednesday | 09:30–10:10 | Explore | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: 1. Name the missing value. 2. Match the units. 3. Choose $M = I \div A$, $I = M \times A$ or $A = I \div M$. 4. Show each line of working.

Work together

Work alone or in a pair. One person chooses the relationship; the other checks the units. Swap for the next item.

1. **A. Image = 20 mm; actual = 0.1 mm. Calculate magnification.**

2. **B. Image = 50 mm; actual = 0.5 mm. Calculate magnification.**

W3L2 | Continue and improve

Use after the core questions when directed

3. **C. Magnification = $\times 400$; actual = 0.05 mm. Calculate image size.**

4. **D. Image = 30 mm; magnification = $\times 600$. Calculate actual size.**

5. **E. Image = 25 mm; actual = 50 μm . Calculate magnification.**

6. **F. Put nm, μm and mm in order from smallest unit to largest.**

Optional transfer

What happens to M if only the printed image length doubles?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W3L3 | Find the first faulty step

Friday | 09:30–10:10 | Do | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Check the units first. Then check the calculation. For a scale bar, count how many bar lengths fit the cell image.

Work together

Read the stated measurements. This route does not depend on the printer scale.

1. A cell image is 48 mm long; the actual cell is 80 μm . Sam writes $48 \div 80 = \times 0.6$. Identify and repair the error.

2. A printed scale bar represents 20 μm . It measures 10 mm on the paper; the cell measures 25 mm. Calculate the cell's actual length.

W3L3 | Continue and improve

Use after the core questions when directed

3. The print is enlarged to 200%. Explain what changes and what stays the same.

Optional transfer

Two pupils judge the same fuzzy image edge differently. Suggest a sensible reporting choice.

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W4L1 | Counter crossings

Lesson 1 of Week 4 | 40 minutes | Introduce | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Net movement = crossings one way minus crossings the other way. Start: “More particles move from ___ to ___ because...”

Work together

Draw two equal-volume regions: 12 particles left and 3 right. Predict the initial net direction.

Illustrative model/sample data; not measurements made by this class.

Model interval	Left → right	Right → left
Early	8	2
At balance	5	5

Your labelled sketch / arrows:

1. Use the early crossing counts to calculate net movement and direction.

2. At balance, how much net movement is there? What is still happening?

W4L1 | Continue and improve

Use after the core questions when directed

3. Explain one reason a higher temperature can increase diffusion rate.

Optional transfer

Why must the two starting regions have equal volumes if particle counts alone are used to compare concentration?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W4L1 | Arrival — choose one route

Four quick questions | Point, say, sign, draw, type or write | You may pass

Choose one route. Point, say, sign, draw, type or write. You may pass or say not sure. Use the screen or paper; you do not need to do both. Nothing here assumes you have met diffusion before. The questions use everyday ideas about smells, drinks and breathing; an adult can read them to you or write your words.

Supported route

1. Someone opens a bottle of perfume at the front. Can people at the back smell it after a while? Yes / No.

2. Two glasses of squash: one strong, one weak. Which one has more squash particles in the same amount of water? Strong / Weak.

3. In a gas, are the particles close together or far apart? Close / Far apart.

4. When you breathe in, where does oxygen get into your blood? Lungs / Stomach.

Standard route

1. A smell spreads across a still room. What is actually moving from the bottle to your nose?

2. What does the word concentration mean when we talk about a solution or a gas?

3. Particles move in solids, liquids and gases. In which state do they move most freely?

4. Name the gas that our blood collects in the lungs and the gas it gives up there.

Stretch route

1. A smell reaches the back of a still room. Explain what makes the particles spread when nothing is pushing them.

2. Two areas have different concentrations of the same substance. What name do we give to that difference?

3. Why do particles in a warm gas spread out faster than particles in a cold gas?

4. Oxygen passes from the air sacs of the lungs into the blood. Suggest why it goes that way and not the other.

W4L1 | Knowledge organiser

Keep this page; use it in any stage

Objective: Explain diffusion and the factors that affect its rate.

Success criteria

1. Define diffusion using net movement and concentration gradient.
2. Name three factors that change the rate of diffusion and say which way each one changes it.
3. Apply diffusion to a real example in the body and to the model shown, keeping the two apart.

Key words

Word	Tier	Meaning
Diffusion	Tier 3	The net movement of particles from a higher concentration to a lower concentration.
Concentration	Tier 2	How many particles of a substance there are in a given volume.
Concentration gradient	Tier 3	The difference in concentration between two places. Particles diffuse down it.
Net movement	Tier 3	The overall movement once movement in both directions has been counted.
Passive	Tier 3	Needing no energy from the cell; diffusion is passive.
Surface area	Tier 2	The total area available for particles to cross.
Membrane	Tier 3	A thin layer that particles must cross, such as a cell membrane.
Alveolus	Tier 3	One of the tiny air sacs in the lungs where oxygen enters the blood. Plural: alveoli.
Rate	Tier 2	How fast something happens; here, how quickly particles move across.

Key ideas

Particles move randomly all the time. Diffusion is the net result of that random movement, not a push.

Three factors raise the rate of diffusion: a steeper concentration gradient, a higher temperature and a larger surface area. A shorter distance to cross also raises it.

Diffusion needs no energy from the cell. Movement against a gradient using energy is active transport, a different process.

In the lungs, oxygen diffuses from the air in an alveolus into the blood; carbon dioxide diffuses the other way. The alveolus wall is one cell thick.

A particle model of diffusion across a membrane. Particles are drawn far larger and fewer than the real thing, and the membrane is drawn as a line. This is an explanatory model, not a measurement of any real gas.

Diffusion or not?

Diffusion: oxygen into the blood at the alveoli; carbon dioxide out of a respiring cell; a smell spreading across a room.

Not diffusion: glucose pumped into a root hair against its gradient (active transport); water across a partially permeable membrane (osmosis, a special case treated in the next lesson but one).

W4L1 | Exit ticket — complete one route, once

Two small checks | You may pass

Two small checks to finish. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

Supported route

1. Diffusion is the net movement of particles from a higher concentration to a ____ concentration.
(lower / higher)

2. A steeper concentration gradient makes diffusion happen _____. (faster / slower)

Standard route

1. Complete the definition: diffusion is the ____ movement of particles from a higher to a lower ____.

2. Name two factors that increase the rate of diffusion.

Stretch route

1. Explain why a thinner membrane increases the rate of diffusion. Use the word distance.

2. Explain why warming a gas increases the rate of diffusion. Use the words energy and particles.

Next lesson: Diffusion in the Lungs: Testing Breath.

Verbal, written, drawn, signed, symbol or choice selected, demonstrated or re-attempted, adult scribed: all count.

W4L2 | Build an alveolus explanation

Wednesday | 09:30–10:10 | Explore | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Use three parts: the feature, what it changes, and how that affects diffusion. Start: “Thin walls help because...”

Work together

Sketch an alveolus next to a capillary. Label air and blood.

Gas	Alveolar air relative to arriving blood	Net direction
Oxygen	Higher	
Carbon dioxide	Lower	

Your labelled sketch / arrows:

1. Complete both net-direction entries.

2. Explain how thin walls help oxygen uptake.

W4L2 | Continue and improve

Use after the core questions when directed

3. Explain how blood flow helps maintain gas exchange.

Optional transfer

If the wall thickens, will extra ventilation necessarily restore the original exchange rate?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W4L3 | Evidence conference

Friday | 09:30–10:10 | Do | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Describe: what changes in the numbers? Explain: why does it change? Start: “When the area doubles, ____ because...”

Work together

Read the controlled comparison below; it is illustrative model data, not a class experiment.

Illustrative model/sample data; not measurements made by this class.

Model patch	Contact area / cm ²	Transfer / model units per minute
A	10	4
B	20	8

1. Describe the comparison using numbers.

2. Concentration difference, membrane and temperature are held constant. Explain the increase.

W4L3 | Continue and improve

Use after the core questions when directed

3. Does this prove that every real medicine patch should be doubled in size? Explain.

Optional transfer

Name a repeat measurement that would strengthen this comparison.

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W5L1 | Membrane decisions

Monday | 09:30–10:10 | Introduce | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Name the substance that moves. Check which solution is more dilute. Add the membrane and net direction. Start: “Water moves...”

Work together

Draw a plant cell in each of three labelled surroundings: more dilute than its cell solution, equal, and more concentrated.

Your labelled sketch / arrows:

More dilute outside	Equal concentrations	More concentrated outside

1. Predict net water movement in the three cases, in order.

2. Explain why the potato may lose mass in a solution more concentrated than its cell solution.

W5L1 | Continue and improve

Use after the core questions when directed

3. If water can mix freely and there is no separating membrane, why is “osmosis” not the precise name?

Optional transfer

At equal concentrations, has water stopped moving?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W5L2 | Design a fair potato investigation

Wednesday | 09:30–10:10 | Explore | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Change = final mass minus initial mass. Percentage change = change ÷ initial mass × 100. Keep the + or - sign.

Work together

Choose the paper/sample route unless the approved practical has been arranged with suitable staffing and equipment.

Illustrative model/sample data; not measurements made by this class.

Solution label	Initial mass / g	Final mass / g	Change / g	Change / %
Water	4.00	4.40		
0.4 mol dm ⁻³ sucrose	4.00	4.00		
0.8 mol dm ⁻³ sucrose	4.00	3.60		

Route actually used: planning/sample data / approved practical

1. These are illustrative sample data. Complete mass change and percentage change.

2. Name the independent variable, dependent variable and two controls.

W5L2 | Continue and improve

Use after the core questions when directed

3. Some cylinders are not blotted. Explain the problem and a specific improvement.

Optional transfer

Would one cylinder at each concentration show the spread of results?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W5L3 | Data detective

Friday | 09:30–10:10 | Do | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Find each mean: add the repeats and divide by their number. Describe the trend. Keep unusual results visible and explain any choice to leave one out.

Work together

Use class data if available; otherwise label this dataset “illustrative sample data”.

Illustrative model/sample data; not measurements made by this class.

Sucrose / mol dm ⁻³	Repeat 1 / %	Repeat 2 / %	Repeat 3 / %	Mean / %
0.0	18	20	22	
0.2	10	12	14	
0.4	0	2	4	
0.6	-10	-8	-6	
0.8	-18	-20	-22	

1. Complete all five means. Describe the trend.

2. Estimate zero change using a straight line between the nearest values either side of zero.

W5L3 | Continue and improve

Use after the core questions when directed

3. An extra repeat at 0.6 gives +9%. What should you do before excluding it?

Optional transfer

Add the +9% repeat at 0.6. Find the new mean and explain its effect.

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W6L1 | Carrier and energy model

Monday | 09:30–10:10 | Introduce | Editable pupil sheet

Name: _____ Date: _____

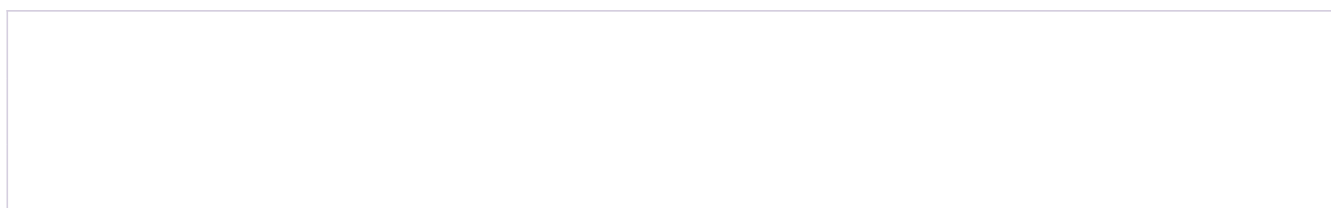
Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Compare the concentration outside and inside. Add an inward arrow. Start: “The ions move from ____ to _____. This needs...”

Work together

Draw two equal-volume regions: 2 nitrate-ion counters outside and 8 inside, separated by a membrane.

Your labelled sketch / arrows:



1. Which way is nitrate moving relative to its concentration gradient?

2. Name the mechanism and its energy source.

W6L1 | Continue and improve

Use after the core questions when directed

3. If respiration is greatly reduced, predict the effect on nitrate uptake and explain.

Optional transfer

Does a large surface area alone prove active transport?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W6L2 | Two-clue uptake investigation

Wednesday | 09:30–10:10 | Explore | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Use two clues: direction compared with the gradient, and what happens when respiration is reduced. Start: “This suggests ____ because...”

Work together

Read the concentration evidence. The numbers are model values in the same units.

Illustrative model/sample data; not measurements made by this class.

Context	Outside	Inside	Normal uptake index	Inhibited index
Root nitrate	2	8	100	15
Gut glucose	3	9	100	20

1. For each context, identify the gradient direction of inward uptake.

2. Explain why the two clues support active transport.

W6L2 | Continue and improve

Use after the core questions when directed

3. Link a root-hair adaptation and a gut-cell adaptation to their functions.

Optional transfer

What extra check would help show that the inhibitor did not simply kill the cells?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W6L3 | Transport card sort

Friday | 09:30–10:10 | Do | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Ask: what moves, which way compared with the gradient, and is cellular energy needed? If a clue is missing, say what more you need to know.

Work together

Cut the four scenario cards or number them on paper; a seated route works equally well.

1. **A. Oxygen has a higher concentration in alveolar air than in arriving blood. It moves into blood.**

2. **B. Water enters a root cell from a solution more dilute than the cell solution, through a partially permeable membrane.**

W6L3 | Continue and improve

Use after the core questions when directed

3. **C. Mineral ions enter a root cell against their gradient; uptake requires energy from respiration.**

4. **D. A substance crosses a cell membrane. No other information is given.**

Optional transfer

Write one similarity and two linked differences between diffusion and active transport.

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

Transport scenario cards

Place, number or point; cutting is optional

Sort into diffusion, osmosis, active transport or need more evidence. Give the deciding clue.

Sort each card. Give the deciding clue before naming the process.

Diffusion	Osmosis	Active transport	Need more evidence
A. Oxygen has a higher concentration in alveolar air than in arriving blood. It moves into blood.		B. Water enters a root cell from a solution more dilute than the cell solution, through a partially permeable membrane.	
C. Mineral ions enter a root cell against their gradient; uptake requires energy from respiration.		D. A substance crosses a cell membrane. No other information is given.	

My deciding clues

W7L1 | Repair the knowledge map

Monday | 09:30–10:10 | Introduce | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Choose two ideas. Link them with “because” or “whereas”. For a cell adaptation, name the feature and what it helps the cell do.

Work together

Teach the three cell adaptations below before using them in the map.

Cell	Feature	Function
Sperm	Tail; many mitochondria	Movement towards the egg; energy for movement
Egg	Nutrient-rich cytoplasm	Supports early development after fertilisation
Ciliated epithelial cell	Beating cilia	Moves mucus and trapped particles along the airway

1. Explain one of the cell adaptations using feature → function.

2. Correct: “A larger image always has more resolution.”

W7L1 | Continue and improve

Use after the core questions when directed

3. Write three reasoned links between magnification, units, osmosis, percentage change, active transport and respiration.

Optional transfer

Choose a review priority and quote one piece of your work as evidence.

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W7L2 | Command-word clinic

Wednesday | 09:30–10:10 | Explore | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Describe = say what happens. Explain = say why. Compare = link a similarity or difference. Calculate = show working. Evaluate = judge using evidence.

Work together

Identify the command word before you answer.

1. **Describe:** concentration rises from 0.2 to 0.6 mol dm⁻³ while mean mass change falls from +12% to -8%. **State the pattern.**

2. **Explain:** “The alveolar wall is thin.” Add the missing mechanism and consequence.

W7L2 | Continue and improve

Use after the core questions when directed

3. **Calculate:** image length is 30 mm and actual length is 75 μm .

4. **Evaluate:** “We should repeat it.” Make this a specific, justified improvement to the potato investigation.

Optional transfer

Compare is not two unrelated lists. Write one linked transport comparison.

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

W7L3 | Independent evidence and response

Friday | 09:30–10:10 | Do | Editable pupil sheet

Name: _____ Date: _____

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

Optional support: Choose one real answer. Underline the part you will improve. Write the change in your own words and say why it is clearer.

Work together

In the 40-minute lesson, complete questions 2, 3 and 5 from the assessment booklet during the 10-minute Independent Task.

1. **My question numbers and first attempt are on the assessment pages. Which answer will I improve?**

2. **What exact scientific detail did I change, and why is it clearer?**

W7L3 | Continue and improve

Use after the core questions when directed

3. What should happen next because of my feedback?

Optional transfer

Explain why an answer can include key words but still be scientifically wrong.

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____